

Display Data: Histograms

Lesson 8-6

Name: _____

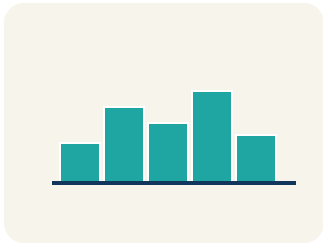
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



Bars side by side: 0-9 pts (3 players), 10-19 pts (8 players), 20-29 pts (4 players)

Histogram

Write the definition:

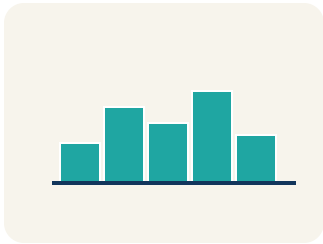


count of one value

If 5 players scored 10-19 points, the frequency for that interval is 5

Frequency

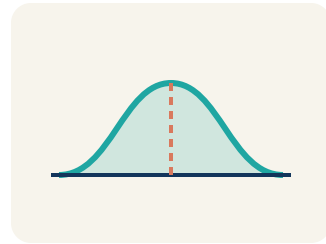
Write the definition:



0-9, 10-19, 20-29 are intervals of width 10 — each covers 10 values

Interval

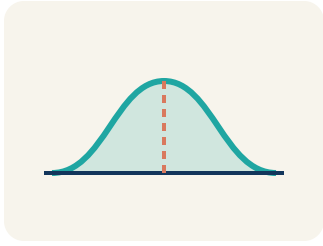
Write the definition:



Most data in the middle with fewer at the ends = bell-shaped; most on one side with a tail = skewed

Distribution

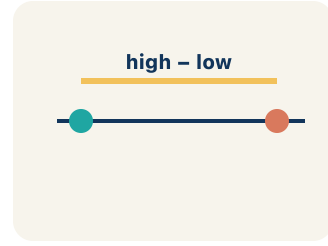
Write the definition:



A histogram that is tallest in the middle and shorter on both sides is symmetric

Data distribution

Write the definition:



Data in just 2 intervals = low variability. Data across 6 intervals = high variability

Variability

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can make and read a histogram to display data in intervals.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Histogram

Frequency

Interval

Distribution

Data distribution

Variability

1

A bar graph that shows how often data falls into equal intervals is a _____.

2

How many data values fall in an interval is the _____.

3

An equal-sized range of values used to group data is an _____.

4

The overall pattern of how data is spread out is the _____.

5

The way data values are spread out is the _____.

6

How much the data values differ is the _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A histogram shows these frequencies: 0–4: 3 players, 5–9: 8 players, 10–14: 12 players, 15–19: 5 players. How many players are represented in total?

- A. 28
- B. 12
- C. 20
- D. 15

Step 1 Add all frequencies: $3 + 8 + 12 + 5 = 28$ players.

 **Answer:** A. 28

 We do — together

Solve this one with your class using the same steps.

Problem: In a histogram, what does the height of each bar represent?

- A. The frequency (count) of data in that interval
- B. The average of the data in that interval
- C. The range of the interval
- D. The median of the data

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: How is a histogram different from a regular bar graph?

- A. Histogram bars touch (no gaps) because the intervals are continuous ranges
- B. Histograms use colors and bar graphs don't
- C. Bar graphs show numbers and histograms show words
- D. There is no difference

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A histogram of steals per game has these frequencies — 0–9: 3 players, 10–19: 9 players, 20–29: 5 players, 30–39: 2 players. Which interval has the highest frequency (tallest bar)?

- A. 10–19 with 9 players
- B. 20–29 with 5 players
- C. 0–9 with 3 players
- D. 30–39 with 2 players

Show your work:

2. Read this histogram of assists per game — 0–4: 2 players, 5–9: 8 players, 10–14: 5 players, 15–19: 1 player. About how many players are on the team?

- A. 16
- B. 8
- C. 4
- D. 15

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A teacher collected test scores and made two different histograms — one with intervals of 5 (50–54, 55–59, etc.) and one with intervals of 20 (50–69, 70–89, 90–109). Both use the same data. How might the two histograms look different? Which interval size gives you more detail about the distribution? When might the larger interval be better?

Sentence starter: The histogram with intervals of 5 would have ____ bars and show _____. The histogram with intervals of 20 would have ____ bars and show _____. Smaller intervals are better when _____, and larger intervals are better when _____.

Show your work:

Reflect — Exit Ticket

A histogram of player heights shows: 60–63 in: 2, 64–67 in: 7, 68–71 in: 9, 72–75 in: 4. Which interval has the most players?

- A. 68–71 inches
- B. 64–67 inches
- C. 72–75 inches
- D. 60–63 inches

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Histogram
2. **Notes 2:** Frequency
3. **Notes 3:** Interval
4. **Notes 4:** Distribution
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **We Try (notes):** A. The frequency (count) of data in that interval — *Each bar's height shows the frequency — how many data values fall within that interval.*
8. **You Try (notes):** A. Histogram bars touch (no gaps) because the intervals are continuous ranges — *In a histogram, bars touch because each interval connects to the next (0–9, 10–19, etc.) — the data is continuous.*
9. **Try It 1:** A. 10–19 with 9 players — *The 10–19 interval has the greatest frequency (9 players), so its bar is the tallest.*
10. **Try It 2:** A. 16 — *Add every bar's frequency: $2 + 8 + 5 + 1 = 16$ players total.*
11. **Exit Ticket:** A. 68–71 inches — *The 68–71 inch interval has the highest frequency of 9 players.*